

HALO: Configuration-Calibrated Evidence Retrieval for Open-Label Human Activity Recognition

ABSTRACT

Human activity recognition systems are commonly tied to one sensor layout and one activity vocabulary. We study a deployment setting in which the device, placement, sampling regime, candidate activity names, and number of labeled demonstrations may all change. HALO represents acquisition configuration instead of normalizing it away. Its tokenizer computes frequency features on each stored sampling grid, defines bands in hertz, records acquisition-limited observability and patch-limited resolution, and folds the axes of each physical sensor into one token. We train the encoder without activity labels, using masked contextual prediction and augmentation consistency.

Classification uses a versioned bank of sensor-patch embeddings. A closed-form evidence path first enforces modality and gravity compatibility, then uses a low-rank text-conditioned gate to decide whether a query and stored sensor can observe each candidate concept. It applies this admissibility test before cosine ranking, votes with enrolled-label identity or corpus-label text, and sums votes across query sensors. Adding a demonstration changes memory but does not update model parameters. The current Phase-A run uses 18 public datasets, 56 acquisition configurations, and 166 canonical labels; we reserve ten disjoint datasets for development and testing. We report the design, its implementation, and the enrollment protocol with its controls. We report no accuracy number: the gate has not been fitted against the completed encoder’s memory, so no result is attributable to the system described here, and the seven-dataset test roster stays sealed.

1 INTRODUCTION

A physiotherapist may ask a patient recovering from knee surgery to perform six exercises at home. The patient already owns a phone and a watch, but the exercise set was selected for that patient and may not appear in a public dataset. Collecting enough annotated examples to train a new model for every device and placement is rarely practical. Most inertial human activity recognition (HAR) systems assume the opposite setting: they fix the sensor layout and activity set, collect an annotated dataset, and fit a classifier for that combination.

We study a setting with three requirements. First, the activity set is *open*: an application supplies its candidate names at inference, including names absent from pretraining. Second, the input is *heterogeneous*: devices differ in placement, modality, channel availability, sampling regime, gravity convention, and filtering. Third, a deployment may provide zero, one, or a few labeled examples of each candidate. These requirements interact. A stored wrist example may be useful for recognizing an arm exercise from another wrist device, but it should not decide an arm gesture observed only by a pocket phone, however close their embeddings happen to be.

Two limitations motivate HALO.

(L1) Acquisition configuration changes what evidence is observable. Rotation, placement, sampling, firmware filtering, and sensor availability jointly change the recorded signal. A larger corpus improves coverage but cannot enumerate their continuous combinations. Nor does resampling make two instruments equivalent: interpolation cannot recover frequencies that the acquisition clock or device filter removed. We therefore expose selected configuration facts to both the representation and the retrieval rule.

(L2) A geometric neighbor is not automatically admissible evidence. Open-label methods often align sensor features with activity-name embeddings and classify by text similarity [46, 56, 84]. Yet language encodes objects and goals that an IMU may not observe, while small wording changes may distinguish a posture from a transition. We consequently compare recordings in sensor space, but condition the right to vote on whether the query and stored sensor configurations can observe the candidate concept. Section 2.3 gives concrete counterexamples without treating author-assigned motion relations as quantitative ground truth.

HALO. HALO has a label-free representation stage and an evidence stage. The tokenizer defines frequency bands in physical hertz on the stored grid, preserves signed low-frequency acceleration, and marks bands unavailable under the acquisition rate or unresolved by the patch duration. It folds the axes of each accelerometer or gyroscope into a sensor token with an explicit axis-validity mask. The token also receives a frozen text description and activity-invariant acquisition statistics.

Phase A learns from masked contextual prediction and consistency across physical augmentations.

At inference we freeze the encoder. A versioned memory stores one row per patch and sensor, with label and provenance. The evidence path applies hard modality/gravity compatibility, a learned low-rank admissibility gate, cosine ranking within the surviving rows, voting, and closed-form sensor fusion. An application appends a labeled demonstration to memory and binds it directly to its own candidate; no gradient update runs. Candidate names enter at run time rather than through a fixed classifier head.

We make four contributions.

- **We design a sensor-granular tokenizer** that defines frequency bands in physical hertz on each stored sampling grid, folds the axes of one physical sensor into a single token with an explicit validity mask, and marks the bands that the acquisition rate or the patch duration leaves unresolved. It states what a configuration *cannot* observe rather than presenting missing information as measured zero.
- **We separate admissibility from similarity** in the retrieval rule. A low-rank text-conditioned gate decides whether a query and a stored sensor can observe a candidate concept, and we apply that test *before* ranking so a rejected row frees its slot instead of merely losing weight.
- **We build a leakage-aware enrollment protocol** in which support and query come from different recorded executions, and we ship five controls with every number: ungated retrieval, support removal, support-label shuffling, prototypes, and ridge heads on the same frozen features.
- **We state what the protocol must contain before it can test the claim.** A two-sided admissibility score collapses to a per-candidate constant whenever the enrolled and queried configurations coincide, so cross-configuration cells are a precondition for measuring the mechanism rather than an extension. We keep the seven-dataset test roster sealed.

2 MOTIVATION

2.1 Application Requirements

Rehabilitation, occupational monitoring, and small field studies motivate an open activity set, several sensor configurations, and useful behavior from few demonstrations. The application, not the pretraining corpus, determines the relevant labels. Inertial traces are also difficult to

label in isolation: annotation commonly requires synchronized video, a scripted protocol, or self-report. A memory-based system can accept a small number of demonstrations without fitting a new model and can keep those recordings local to the deployment.

2.2 Acquisition Configuration Determines Observability

The recorded signal depends jointly on mounting orientation, body placement, sampling rate, device-side filtering, gravity convention, and sensor availability. Some differences are explicit in metadata; others are properties of the conversion. For example, one source may expose total acceleration while another exposes gravity-removed acceleration. Upsampling changes an array’s length but cannot restore bandwidth absent at acquisition. Likewise, full rotation invariance can discard posture information carried by gravity.

These facts motivate a distinction between *compatibility*, *admissibility*, and *similarity*. Compatibility asks whether two rows have comparable modalities and gravity states. Admissibility asks whether both sensor configurations can observe the candidate concept. A pocket accelerometer can be relevant to gait but may be uninformative for a fine arm gesture. Similarity ranks the remaining evidence. Applying these operations in the opposite order permits a geometrically close but physically irrelevant row to occupy a limited retrieval slot.

HALO therefore represents three kinds of acquisition information. Frequency bands are defined in hertz and carry masks for acquisition bandwidth and patch resolution. One frozen text descriptor states device, modality, placement, gravity convention, and partner-sensor presence. A separate activity-invariant vector measures acquisition behavior such as gravity magnitude, noise floor, quantization, clipping, rate fidelity, and rest bias. Missing measurements have explicit support bits. This split permits the paper to test whether the representation learned motion or merely recognized the source dataset.

2.3 Activity-Name Similarity Is Not a Motion Metric

An open-label model cannot rely on one fixed output head. A common alternative embeds candidate names and aligns sensor features with the same sentence space [46, 56, 84, 90]. That space reflects linguistic relation, not only observable motion. *Drinking coffee* and *drinking water* differ in object semantics while a wrist sensor may observe nearly the same gesture. Conversely, *sitting* and *sitting down* have similar words but denote a

Table 1: The active mechanisms and the distinction between each preserves.

Problem	HALO mechanism
Different sampling regimes	Gaussian filterbank with logarithmic, constant- Q -inspired centers defined in hertz; acquisition observability and patch-resolution flags
Missing or damaged axes	Sensor folding with a three-bit axis-validity vector
Declared vs. measured configuration	Frozen sensor text plus separately measured, activity-invariant acquisition statistics
Irrelevant geometric neighbors	Compatibility and candidate-specific admissibility before top- k cosine ranking
Deployment-specific labels	Identity-bound enrolled votes and append-only memory

Figure 1: Activity-name similarity is not a motion metric. Qualitative counterexamples only; no author-assigned motion distance is reported as a measurement.

posture and a transition. Figure 1 presents these as conceptual counterexamples; it does not assign quantitative motion distances by author judgment.

HALO uses language in two restricted roles: to describe acquisition configuration and to name candidates or corpus labels. It does not train the encoder to reproduce sentence distances. Stored recordings are ranked in sensor-feature space, while configuration-conditioned admissibility decides whether a neighbor may vote for a candidate. An enrolled demonstration is bound to the candidate by identity, so arbitrary names remain usable when labeled examples are present.

Dependency among components. Frequency coordinates and masks make rows from different sampling regimes interpretable. Sensor folding and descriptors expose configuration to the encoder. The evidence engine then uses that configuration to reject or downweight rows whose placement cannot observe a candidate. The thesis is therefore not that each component is individually novel, but that acquisition metadata calibrates evidence rather than being normalized away.

3 DESIGN OF HALO

3.1 System Overview

HALO separates representation learning from evidence-based classification (Figure 2). Phase A trains a sensor encoder without activity labels. The encoder is then frozen and used to build a versioned memory. At inference, a closed-form evidence path compares query and memory rows. The only fitted component in this path is a small admissibility gate that maps a sensor description and a candidate concept to one scalar. There is no relational decoder, learned retrieval projection, or deployment-specific output head in the active system.

For a window containing an accelerometer and gyroscope, the spectral front end first produces features for each scalar axis. The three axes of each modality are then folded into one token per patch and physical sensor. Frozen text describes what the sensor was declared to be, while a separate vector records activity-invariant acquisition measurements. A transformer contextualizes these tokens over physical time and across sensors. The

memory stores the resulting sensor rows with labels and provenance.

3.2 Sensor-Granular Tokenizer

3.2.1 Frequency features in physical units. For each channel patch, the tokenizer removes the mean after applying a validity mask, applies a Hann window, and computes a zero-padded real DFT of size $S=256$. If the stored array rate is r , bin m represents mr/S hertz. It then evaluates $K=32$ Gaussian filters whose centers are logarithmically spaced from 0.3 to 15 Hz and whose nominal bandwidth scales with center frequency ($Q=4$). The construction is *constant- Q -inspired*: the cited constant- Q literature supports geometric center spacing [11, 12], while the Gaussian weighting and its parameters are HALO design choices.

The tokenizer operates on the stored grid and avoids an additional interpolation step in the model path. When a converter has already resampled a device stream, the stored rate still defines the DFT coordinate, but the original acquisition rate bounds observability. The representation therefore does not assert that different devices have identical filter responses; it only assigns the same physical coordinate to corresponding bands.

For center c_k and width $\sigma_k = c_k/(2Q)$, the observability mask is

$$o_k = \mathbf{1}[c_k + 2\sigma_k \leq 0.9r_{\text{source}}/2].$$

The patch-resolution flag is the clipped number of cycles in duration D , $q_k = \min(1, c_k D)$. This is a heuristic indication of spectral support, not a theorem that one cycle is sufficient. Unobservable bands and observed zero energy remain distinguishable.

We retain the mean removed from the spectral path as a signed per-axis low-frequency feature for accelerometers. It can carry gravity and posture information. We do not attach the numerical posture-separation claims from an earlier draft because their generating artifact is

[FIGURE PLACEHOLDER]

End-to-end pipeline: Phase-A label-free encoder → frozen per-patch, per-sensor bank → compatibility, admissibility, ranking, voting, sensor fusion.

Figure 2: HALO end to end. Phase A learns one sensor representation without activity labels. A versioned bank stores frozen per-patch, per-sensor rows. Classification applies hard compatibility, candidate-conditioned admissibility, cosine ranking, voting, and sensor fusion. Enrollment appends labeled rows; it does not update parameters.

absent; the ablation in Section 5.3 is the evidential test of this design choice.

3.2.2 Folding axes into physical sensors. Each accelerometer or gyroscope is a three-axis physical sensor. HALO concatenates the three per-axis patch tokens in fixed x, y, z order, zeros invalid axes, appends a three-bit validity vector, and projects the result to one token. Thus “axis reads zero” differs from “axis absent.” A six-channel accelerometer–gyroscope stream becomes two tokens per patch, while an accelerometer-only stream becomes one. Axis-role text is unnecessary after folding because axis identity is represented by its position inside the sensor token.

The augmentation stack respects this structure. We apply one $SO(3)$ rotation jointly to a triad and its co-located gyroscope, and we permit sensor masking only when another modality at the same placement remains visible, so the model never has to reconstruct a wrist from an ankle or a pocket.

3.2.3 Declared descriptors and measured acquisition bias. Each sensor receives two frozen artifacts. A 384-dimensional sentence embedding describes device, modality, placement, gravity state, and whether a partner modality was present. A 14-dimensional measurement vector contains seven standardized statistics and seven support bits: gravity magnitude, gravity presence, high-band noise floor, quantization step, clipping fraction, native-to- stored rate fidelity, and rest bias. These statistics are estimated on training subjects and, where possible, quiescent windows; unsupported fields remain unsupported rather than being imputed as measurements.

Separate learned MLPs project the two artifacts into the token width and inject them through gated residuals. Keeping the artifacts frozen preserves their audit meaning. The representation may still encode dataset provenance, so we measure dataset-identification accuracy and keep similarity in the acquisition-statistics vector disabled in the active retriever unless a provenance guard shows it is safe.

3.2.4 Context encoder. The transformer encoder [73] has width 256, six layers, eight heads, and a frozen all-MiniLM-L6-v2 text encoder [57]. Attention alternates over time within a sensor and across sensors at a patch; padding and sensor masks exclude missing values. Rotary positions use elapsed seconds rather than token indices [70]. Each window is represented on one fixed one-second patch grid, and pooling weights each patch by its real temporal support, so a final short patch counts for less than a full one. A multi-resolution variant remains available as an ablation.

3.3 Phase A: Label-Free Pretraining

Phase A has two top-level objectives. Their coefficients are calibrated once at step 2,000 from encoder-gradient norms and then fixed.

Masked contextual prediction. A student predicts contextual sensor tokens produced by an exponential-moving-average teacher [2, 3, 26]. Masking includes contiguous time blocks and, when physically well posed, a complete modality at one placement. A descriptor-mask variant, which hides a sensor’s text and scores an InfoNCE retrieval of the true frozen descriptor [72], is implemented but carries zero weight in the reported recipe.

Augmentation consistency. VICReg [6] aligns two augmentations of the same window while maintaining feature variance and reducing covariance. The reference recipe applies one transform: an $SO(3)$ mounting rotation drawn independently for each view. It avoids declaring other windows to be negatives when they may contain the same activity. Activity labels are not used in either gradient path. Checkpoint selection uses subject-disjoint k -nearest-neighbor balanced accuracy on the Phase-A validation subjects.

3.4 Admissibility-Gated Evidence

Bank. The current schema stores one row per patch and sensor: a 256-dimensional feature, the frozen sensor descriptor, acquisition measurements, label, subject, dataset, configuration, recording event, time, and resolution. We bind the bank to the exact encoder checkpoint

[FIGURE PLACEHOLDER]

Evidence path: compatibility and admissibility applied before top- k ranking

Figure 3: Active evidence path. Compatibility and admissibility are applied before candidate-specific top- k ranking. Enrolled rows vote only for the candidate to which the application bound them; corpus rows vote through label-text similarity. Per-sensor votes are summed.

and schema, and exclude query-window, event, and subject identifiers wherever the evaluation protocol demands it. Enrollment adds rows whose episode-local candidate identity is explicit.

Train-only resolvability measurement. For every training sensor and concept with adequate support, a subject-disjoint one-versus-rest k NN probe measures how well that sensor distinguishes the concept. Chance balanced accuracy maps to zero. We measure accelerometers and gyroscopes separately and report simultaneous-placement contrasts separately. Activity strings only group the measurement cells; this probe never embeds them.

Admissibility gate. A rank- r bilinear model maps frozen sensor and concept text to one weight:

$$u = Ae_s, \quad v = Be_c, \quad g(s, c) = \sigma(u^\top \Lambda v + b),$$

where $e_s, e_c \in \mathbb{R}^{384}$ and $r=2$ in the active artifact. The low rank limits the gate’s ability to memorize literal stream-label cells. We fit it only to the train-only resolvability table, and for text unlike the fitted sensor or concept supports we blend the prediction toward the table’s neutral median rather than extrapolating confidently. Characterizing generalization requires held-out configuration, dataset, and concept folds; the fitted table itself is not validation evidence.

Prediction. For each query sensor, HALO performs the following operations in order.

- (1) Hard compatibility removes rows with incompatible modality or gravity state. This step is label independent.
- (2) The gate scores the query sensor and memory sensor against each candidate. Their geometric mean requires both sides to observe the concept. Rows below the veto threshold are removed *before* top- k , so a rejected row frees its slot.
- (3) Feature cosine ranks the surviving rows separately for each candidate. The active budget is 64 rows per query patch and candidate. Similarity in the

acquisition-statistics vector has weight zero in the reported artifact.

- (4) An enrolled row votes for its bound candidate by identity. A corpus row votes through the rectified cosine between its stored label text and the candidate text; this is the only $k=0$ bridge.
- (5) Votes are summed across the query’s sensors. This fusion is closed form and does not infer one placement’s signal from another.

Setting every admissibility value to one gives the ungated identity-retrieval control. Prototype and ridge classifiers use the same frozen representations and enrolled examples. Support removal and cyclic support-label shuffling test whether an apparent enrollment gain actually uses the labeled rows.

Scope. The active system does not include an unknown output, confidence calibration, Core ML export, an iOS application, or a public `predict()` API. Selective classification would require truth-absent episodes and risk-coverage evaluation [23]; evidential deep learning is a distinct uncertainty method [60] and is not evidence for the present retrieval rule.

4 IMPLEMENTATION

We implement HALO in PyTorch as a single repository covering conversion, pretraining, memory construction, and the evidence rule. The system is deliberately small where it matters: the trunk holds 7.70 M parameters and the only component fitted after Phase A is a rank-2 bilinear gate with roughly 1.5 k parameters.

4.1 Codebase

Table 2 lists the components. We write one converter per dataset — 35 in total — rather than a generic ingest path, because every public corpus differs in column order, unit convention, gravity handling, and subject encoding, and a generic reader hides exactly the errors that later corrupt a physical feature. Each converter emits whole-recording sessions; the grid builder owns all windowing, so no converter can silently choose its own window policy.

4.2 Training and Memory Cost

We train on one NVIDIA RTX 4090 (24 GB). The reported Phase-A run completes 30,000 steps at batch 256 in 33 minutes of wall clock and peaks at 4.36 GiB of allocated device memory, so a full pretraining run fits comfortably on a single consumer GPU. Mixed-precision training is enabled and we compile only the transformer; compiling the encoder specializes on each batch’s changing unique-text cardinality and runs slower.

Table 2: Implementation components. Line counts exclude generated artifacts.

Component	Lines	Contents
Dataset converters	11.8 k	35 per-dataset readers, unit and gravity normalization
Corpus pipeline	6.9 k	curation policy, windowing, three grid alignments, quality screens
Model	3.9 k	filterbank, sensor folding, conditioning, dual-branch trunk
Phase-A training	7.6 k	sampler, augmentation, masking, objectives, telemetry
Evidence stage	14.3 k	bank build, resolvability, gate, prediction rule, protocol
Evaluation harness	2.1 k	baseline adapters, scoring, protocol fingerprinting
Tests	9.3 k	578 tests across 45 files

Memory construction is the expensive artifact. The schema-4 bank stores one row per patch and sensor, which is 7.65 M rows for a 250,000-window archive and 6.9 GB on disk in half precision. Two engineering choices keep the evidence path tractable. First, we do not store a 384-dimensional sensor descriptor on every row: roughly 10^5 rows are drawn from about 40 distinct sensor descriptions, so we store a de-duplicated table and an index, evaluate the gate once per distinct description, and gather. This is exact, not approximate, because the gate reads only text. Second, we cache stream metadata and text encodings per configuration, so sentence-embedding inference never runs inside a query loop.

4.3 Artifact Binding

A retrieval system built on a cached bank fails in a specific way: an encoder is retrained, the bank is not rebuilt, and every downstream number silently describes a stale embedding space. We make that failure impossible to reach rather than easy to notice.

Every artifact carries a fingerprint of what produced it, and every consumer checks it before doing any work. The bank records the encoder checkpoint hash, the label vocabulary, and behavioral probes of both the embedding path and the sentence-embedding path; loading it re-runs those probes against the live encoder and refuses a mismatch. The gate records the resolvability table hash, the checkpoint that produced it, and the bank it is bound to. The evaluation protocol is itself fingerprinted over the source files that define scoring behavior, so a change to the protocol invalidates cached results instead of blending two protocols into one table. The quality screens are loaded with the same discipline: a missing or stale exclusion cache is a hard failure, because silently re-admitting byte-identical stale-buffer windows would corrupt both training and scoring.

These guards are load-bearing rather than decorative. During development they rejected a legacy resolvability table that had pooled evaluation datasets into a training-only measurement, and they currently refuse every gate artifact fitted before the completed encoder, which is why this paper reports no enrollment result rather than one carried over from a superseded checkpoint.

4.4 Reproducibility

Every quantitative statement in this paper is bound to a named artifact through a checked-in evidence ledger that records the producing command, the source-data manifest, the seed, the checkpoint fingerprint, and the raw output. Values without that chain are not printed as results, which is why several quantities from an earlier draft of this work do not appear here. We will release the converters, the corpus pipeline, the model, the evidence stage, and the evaluation harness.

5 EVALUATION

This section states the corpus the system is trained on, the enrollment protocol and its controls, and the measurements that must exist before we make a performance claim. We report no accuracy number. The Phase-A encoder is complete and its memory and resolvability artifacts are built against it, but the admissibility gate has not been fitted to those artifacts, so no enrollment result is attributable to the system this paper describes.

5.1 Data and Artifacts

Phase A uses 18 public datasets and 56 acquisition configurations. Their materialized native-grid manifests contain 1,963,606 stream-window rows before quality screens and the subject split. The bank built from the completed encoder contains 250,000 balanced source windows, 546 training subjects, all 166 canonical training labels, 1,350,834 patch rows, and 2,629,972 sensor rows. Ten datasets are disjoint from Phase-A training: three are the development roster and seven are sealed for testing. Table 3 gives the full grid-derived roster; its machine-readable provenance is listed in the paper-source evidence ledger.

The completed run trains for 30,000 steps and we select step 27,000 by a subject-disjoint Phase-A validation metric that never sees an evaluation dataset.

5.2 Evaluation Protocol

An evaluation dataset supplies its own candidate inventory; this is a HALO protocol choice, not a claim established by the generalized zero-shot-learning benchmark. Following that benchmark, we separately report whether

Table 3: Current Phase-A and external-evaluation rosters. *Cfg* counts materialized sensor configurations; *rows* counts stream-window rows before the current quality exclusions and subject split, so simultaneous placements are counted separately. [†]The XRF ear stream was acquired at 25 Hz and stored at 50 Hz. [‡]MM-Fit devices have different acquisition clocks and are placed on a shared 100 Hz grid; source rates still bound observability. The seven test datasets have not been used for method selection.

Phase-A training				External evaluation			
Dataset	Hz	Cfg	rows	Dataset (role)	Hz	Cfg	rows
CAPTURE-24 [15]	100	1	1,543,573	MotionSense [43] (dev)	50	1	4,534
WISDM [77]	20	2	72,227	RealWorld [71] (dev)	50	1	11,237
RealDISP [4]	50	9	63,432	Shoaib [62] (dev)	50	1	2,100
XRF V2 [36]	25/50 [†]	6	59,442	InclusiveHAR [30] (test)	50	1	1,260
DSADS [7]	25	5	38,175	USC-HAD [88] (test)	100	1	4,350
NFI-FARED [45]	100	2	29,184	TNDA-HAR [85] (test)	50	1	3,353
PhyTMO [22]	100	8	29,180	UT-Complex [63] (test)	50	1	3,900
Opportunity [16]	30	5	28,785	MoniPar [64] (test)	50	1	12,079
HARMES [13]	50	1	25,583	SPAR [14] (test)	50	2	1,898
FORTH-TRACE [32]	51.2	5	12,330	Upper Limb Use [18] (test)	50	4	1,124
UniMiB SHAR [47]	50	1	11,771				
HHAR [68]	50	1	11,587				
KU-HAR [65]	100	1	11,263				
UCI-HAR [1]	50	1	10,299				
MM-Fit [69]	100 [‡]	4	8,496				
SP-SW-HAR [44]	100	2	3,757				
PAMAP2 [58]	100	1	3,292				
MHEALTH [5]	50	1	1,230				

a candidate was seen or unseen during training [78]. Macro-F1 is primary because activity frequencies are imbalanced.

For enrollment, support and query executions are disjoint. The development evaluation uses cross-subject cohorts over MotionSense, RealWorld, and Shoaib. At each $k \in \{1, 2, 4, 8\}$, we retain the same queries and candidate inventory and nest the support prefixes. A verified recording execution, not an adjacent window, counts as one example. We also run support removal, cyclic support-label shuffling, an all-one admissibility control, prototypes, and ridge heads on the identical features and support split. These controls are part of the claim, not optional ablations: the all-one arm isolates the gate, and the prototype and ridge arms establish what the frozen representation already supports without any retrieval machinery.

The protocol must include cells in which the enrolled configuration differs from the query configuration. A two-sided admissibility test scores the query sensor and the evidence sensor against each candidate; when both are the same sensor, that product reduces to a per-candidate constant and cannot express configuration dependence at all. Cross-configuration cells are therefore not an extension of the protocol but a precondition for it to measure the mechanism.

The test roster is fixed: InclusiveHAR, USC-HAD, TNDA-HAR, UT-Complex, MoniPar, SPAR, and Upper Limb Use. TNDA-HAR has no usable subject-disjoint

enrollment cohort and contributes only to a future zero-support cell. SPAR’s repetitions occur within one bout and do not establish across-session enrollment. MoniPar is the held-out source that can support across-week enrollment. We consume none of these datasets in the present revision.

5.3 Required Measurements Before a Performance Claim

The following experiments are part of the protocol and do not yet have provenance-complete artifacts:

- development enrollment curves for the gate against every control above, on a bank and gate bound to the completed encoder;
- cross-configuration enrollment on simultaneous training streams, reported separately from the external test result;
- held-out concept, dataset, and body-region studies for the gate, including gate-collapse, provenance-capture, and latent-to-measurement guards;
- native-grid versus common-rate, sensor-folded versus channel-level, descriptor-on, acquisition-bias-on, observability-off, and signed-low-frequency ablations;
- all-pairs activity-text similarity versus sensor-centroid similarity, replacing any hand-selected quantitative geometry claim;
- data-matched and released-checkpoint baselines, including the published short fine-tuning baseline

the wearable-foundation-model literature requires; and

- only after the development criteria are met, one evaluation of the seven-dataset sealed test roster with confidence intervals and seen/unseen candidate breakdowns.

We claim no on-device latency, Core ML, iOS, or unknown-class result, because the corresponding implementation and measurement artifacts do not exist in the repository.

6 RELATED WORK

Cross-dataset and foundation models for wearable sensing. DeepConvLSTM [53] is representative of supervised HAR models trained for one benchmark. Later work addresses selected sources of variation [54, 55, 79] or pretrains transferable encoders. LiMU-BERT [80] uses masked reconstruction at 20 Hz, CrossHAR [27] adds a contrastive objective, and ssl-wearables [86] scales self-supervision on wrist accelerometry. Scaling Wearable Models [49] studies population-scale pretraining and imputation; LSM-2 [82] specifically targets incomplete wearable observations. RelCon [81] learns relative motion similarity from roughly one billion sensor segments, and SensorLM [91] pairs wearable signals with language. Tartan IMU [92] studies inertial positioning, while general time-series architectures include MOMENT and PatchTST [24, 51]. Mitra is not included in this group because it is a tabular, rather than time-series, foundation model [89].

The controlled Inertia-1 study varies modality, placement, rate, and window length [83]; the recent HAR foundation-model survey provides a broader taxonomy [8]. We cite the study for its controlled findings and the survey only for the landscape. HALO’s narrower question is whether acquisition configuration can calibrate which retrieved evidence is allowed to vote.

Sampling and acquisition context. Rate-invariant autoencoding disentangles order-preserving temporal warp into rate parameters and a rate-invariant latent [34]. FlowState combines a state-space model with a functional basis decoder and dynamic timescale adjustment for sampling-rate-equivariant forecasting [25]. Universal Spectral Tokenization accepts native wavelength grids in astronomy and proposes extension to other sampled domains; it is adjacent inspiration, not evidence on wearable HAR [61]. MoPFormer learns IMU motion primitives after resampling its datasets to 100 Hz [87]. These systems therefore should not be summarized as sharing one rate-normalization strategy.

Other work supplies acquisition context explicitly. GOAT [46] uses activity and body-position text, ZeroHAR [17] uses categorical sensor context, and CHARM was introduced as a preprint and subsequently presented in the peer-reviewed SensorsVoice work [19, 20]. SensorLLM [39] and oneHAR [76] also use language for sensor context, whereas UniMTS [90] represents placement as a graph and uses rotation augmentation. HALO does not claim that sensor descriptions are new. Its contribution is to separate a declared description from measured acquisition statistics and then use configuration to calibrate retrieval admissibility.

Retrieval and label-efficient recognition. Template matching has long been used for gestures [29, 40, 50]. Matching Networks and Prototypical Networks learn or use comparison rules under episodic tasks [67, 75]. Language and vision systems augment parametric models with external non-parametric memory [10, 28, 33, 37, 48]; not all of these systems keep the parametric model frozen. Memory-modular classification studies adding classes by changing memory contents [31].

In sensing, RAG-HAR retrieves signal descriptors for an LLM [66], and ZARA is a training-free, evidence-grounded LLM agent [38]. TimEE and TIC-FM perform zero- or few-example time-series classification in context [21, 35]. These are not evidence for a target-example k curve without parameter updates unless such a curve is explicitly reported. Appendix A therefore separates open-label evaluation, use of target examples, and no-update enrollment instead of collapsing them into one checkmark.

7 DISCUSSION AND LIMITATIONS

Same-configuration cells cannot test admissibility. The gate scores the query sensor and the evidence sensor against each candidate and combines them. When enrollment and query come from the same configuration, both factors read the same description, and the product reduces to a function of the candidate alone. A query-independent per-candidate constant multiplying an evidence-derived vote can shift an argmax, but only toward whichever concept the gate considers a priori more observable, regardless of what the retrieved evidence says. This is a property of the rule, not of any particular fit, and it is why Section 5.2 treats cross-configuration cells as a precondition rather than an extension. It also means an evaluation restricted to same-configuration cells can measure the mechanism’s cost while being structurally unable to measure its benefit.

The veto threshold is not tied to the distribution it gates. Admissibility below a fixed constant

removes a row from a candidate’s pool. That constant does not move with the fitted gate, whose output scale depends on the encoder that produced the resolvability table. Two failure modes follow. If the neutral value that unfamiliar text falls back to sits above the threshold, abstention rescues every unfamiliar configuration and the veto never fires, leaving only reweighting. If it sits below, an unfamiliar configuration is vetoed on every candidate and contributes nothing. Nothing in the current design asserts an ordering between the two. Expressing the threshold as a quantile of the fitted values, and asserting that ordering at fit time, would remove the failure mode; we have not done so.

Provenance is confounded with hardware in public corpora. Each public dataset generally uses one device model and one preprocessing chain, so a descriptor of acquisition behavior is also a descriptor of which study a recording came from. We respond by keeping the acquisition-statistics similarity term disabled in the reported retriever and by measuring same-dataset retrieval directly rather than assuming it is benign. The confound is a property of the available data and no estimator resolves it. Settling it needs a corpus with the same hardware across studies, or the same study across hardware.

Scope of the recognition problem. HALO classifies presegmented single-activity windows. It has no unknown output and does not perform continuous segmentation or calibrated rejection, so a deployment that encounters an activity outside its candidate set will still name one. Selective classification would require truth-absent episodes and risk-coverage evaluation [23]; we treat it as future work rather than claiming it, because the evaluation artifacts that would support such a claim do not exist.

No accuracy and no on-device claim. We report the design, the implementation, and the protocol, and no recognition accuracy. The encoder is complete and its memory is built, but the gate has not been fitted against that memory, so any number we printed would describe a different system than the one specified here. We likewise make no claim about phone-side latency, energy, or export, because we have not built that path. The design does bound what such a claim would look like: the encoder is 7.70 M parameters and enrollment appends rows rather than running a gradient step, so the runtime question is retrieval over a bank rather than on-device training. The bank is the obvious pressure point — 6.9 GB is a server-side artifact, and a phone-side deployment would need either a per-user subset or a compressed index.

Sealing the test roster. Seven evaluation datasets remain unconsumed, and Section 5.3 states the criteria for unsealing them. This is a deliberate cost: it means this paper reports no external performance number. We think it is the correct trade while the development criteria are unmet, because a test number produced under a protocol that cannot express the mechanism would not be evidence for the claim the paper makes.

8 CONCLUSION

HALO tests one proposition: acquisition configuration should calibrate evidence rather than be normalized away. We build a sensor-granular encoder that defines frequency bands in physical hertz, marks what a configuration cannot resolve, and learns without activity labels; we classify by a transparent sequence of compatibility, admissibility, cosine ranking, voting, and sensor fusion; and we enroll a new activity by appending rows to memory rather than updating parameters.

The proposition remains open, and we are explicit about why. The encoder and its memory are built, but the admissibility gate is not yet fitted against them, so we report no recognition accuracy. The more useful finding is structural: a two-sided admissibility score degenerates to a per-candidate constant whenever enrollment and query share a configuration, so the obvious evaluation — enroll and query on the same device — cannot express the mechanism at all. The next measurement follows directly and needs no retraining: enroll on one device, query from another.

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A SCOPE OF THE CLOSEST SYSTEMS

Table 4 keeps four questions separate: handling acquisition rate, receiving placement or channel context, accepting run-time activity names, and evaluating multiple target example counts without parameter updates. A zero-shot result is not marked as a k curve, and a linear probe or fine-tuning result is not marked as no-update enrollment.

B CORPUS CONSTRUCTION

Table 3 is generated from each stream’s `grids/native/*meta.json`. Window counts are properties of the HALO converters and grid policy, not quantities claimed by the source dataset papers. The evidence ledger records the aggregation command and distinguishes materialized rows, quality-screened rows, subject splits, and the balanced 250,000-window bank.

[FIGURE PLACEHOLDER]

Materialized configurations by stored grid rate (20–100 Hz): 56 Phase-A, 14 external.

Figure 4: Materialized configurations by stored grid rate. The current roster contains 56 Phase-A configurations and 14 external-evaluation configurations. Source acquisition rate is stored separately and bounds observability when a converter has already resampled a stream.

The expanded training roster deliberately includes body-worn stress configurations from DSADS and Opportunity; an earlier draft incorrectly listed both as excluded. Six newly added training sources contribute most of the placement coverage: DSADS, FORTH-TRACE, Opportunity, RealDISP, MM-Fit, and PhyTMO. MoniPar, SPAR, and Upper Limb Use are held out.

Sources not used in the named recipe. HAPT [59] reuses the UCI-HAR participants and recordings, so it is excluded to avoid near-duplicate subjects. RecGym [9] distributes per-axis min-max-normalized data that no longer preserve the physical scale used by the tokenizer. The MobiAct [74] download did not produce a materialized grid and is not counted. HARTH [41] is converted but has no materialized grid. ExtraSensory, NHANES, HMOG, and KneE-PAD are buildable optional scale and stress sources that must be requested explicitly, and are not members of the reported 18-dataset run.

C BASELINE CONTRACTS

Models receive different input information, so a future comparison must disclose both weight source and input contract. LiMU-BERT and CrossHAR require corpus-matched retraining. Released harnet, UniMTS, and NormWear weights were trained on different corpora. ConSE [52] can translate a fixed training classifier into candidate text scores, but it does not make the underlying encoder open label or provide an enrollment mechanism.

D TOKENIZER CHECKS REQUIRED FOR PUBLICATION

The implementation contains unit guards for DFT overflow, acquisition-rate observability, sensor folding, and axis validity. Those software tests do not by themselves establish empirical rate-equivalence or posture utility. The retired figures used hard-coded values without an archived generator. Their replacements must be produced by a checked-in script that records input windows, anti-alias filtering, source and stored rates, checkpoint hash, random seed, and raw feature arrays. The signed-low-frequency ablation must use the same evaluation cohort and report uncertainty.

E PHASE-A SPECIFICATION

The reported run uses a fixed filterbank, sensor-granularity tokens, width 256, six layers, eight heads, batch size 256, 30,000 steps, learning rate 3×10^{-4} , 1,000 warmup steps, weight decay 0.05, gradient clipping at 1.0, and seed 20260718. The two top-level loss coefficients are fixed after calibration at step 2,000: 7.053501 for the JEPA family and 0.653232 for VICReg. The teacher decay is 0.996. Descriptor retrieval carries coefficient zero and is not trained.

Patches are one second on a single resolution. A dataset with n windows is sampled proportional to $n^{0.25}$ with a maximum share of 25%; subjects are sampled proportional to the square root of their window count.

Table 4: Reported scope of related systems. Entries summarize the cited evaluation rather than infer unreported capabilities. “Task fit” includes a probe or fine-tuning; “zero only” is not a target-example curve.

System	Rate/input treatment	Configuration context	Label interface	Target adaptation
LiMU-BERT [80]	common 20 Hz	none	fixed downstream	task fit
CrossHAR [27]	common-rate inputs	none	fixed downstream	task fit
harnet [86]	wrist contract	wrist accelerometer	fixed downstream	task fit
LSM / LSM-2 [49, 82]	multimodal, incomplete data	modality availability	fixed tasks	probe/fine-tune
UniMTS [90]	resampled inputs	placement graph	text-conditioned tasks	task fit
NormWear [42]	common-rate input	variable channels	text interface	zero/task evaluation
ZeroHAR [17]	configured input	categorical context	run-time names	zero only
CHARM / SensorsVoice [19, 20]	channel-tokenized	channel descriptions	task dependent	task evaluation
MoPFormer [87]	all data at 100 Hz	wearable context	fixed tasks	task fit
FlowState [25]	rate equivariant	sampling rate	forecasting output	not HAR
RAG-HAR [66]	retrieved descriptors	sensor evidence	run-time names	zero only
ZARA [38]	multi-sensor evidence	sensor evidence	run-time names	zero only
SensorLM [91]	wearable pretraining	sensor/task prompts	task dependent	task adaptation
HALO	stored-grid bands in Hz	per-sensor text + measurements	run-time names	$k = 0, 1, 2, 4, 8$, no update

Table 5: Planned baseline groups. Results are not reported until every row has an artifact using the same support and query cohorts.

Model	Weight source	Input contract	Candidate bridge
LiMU-BERT [80]	corpus matched	20 Hz, padded 6-channel	ConSE
CrossHAR [27]	corpus matched	common rate, padded channels	ConSE
harnet [86]	released	30 Hz wrist acceleration	ConSE
UniMTS [90]	released	placement graph	native interface
NormWear [42]	released	model-specific common rate	text interface
HALO	current run	per-sensor stored-grid input	evidence rule

Activity labels do not enter the training sampler or loss. The selected checkpoint is step 27,000, which maximizes the predefined subject-disjoint label-stream-macro validation score over the completed run.

F EVIDENCE ARTIFACT AND CONTROLS

The active bank has 250,000 source windows, 56 configurations, and all 166 canonical labels. It records 1,350,834 valid patch rows and 2,629,972 per-sensor rows, and it is bound to the completed encoder checkpoint and to the sentence-embedding path used to build it. A gate artifact must additionally record the resolvability table and bank it was fitted against. Evaluation refuses legacy banks or foreign artifacts unless the mismatch is explicitly acknowledged.

The prediction budget is 64 evidence rows per query patch and candidate. Sensor-bias similarity is disabled.

For semantically unfamiliar sensor or concept text, the gate blends toward a neutral value. The following telemetry is mandatory whenever the gate is reported:

- gate mean, spread, extrema, and vetoed-row fraction;
- same-dataset retrieval with the acquisition-bias blend both off and on;
- held-out concept, dataset, and body-region errors relative to constant and per-concept baselines; and
- within-concept correlation between gate latents and the train-only resolvability table.

Every enrollment result shares one deterministic query cohort and nested support prefixes across the gate, all-one admissibility, support removal, support-label shuffling, prototype, and ridge controls. These controls are part of the claim, not optional ablations.

G EVIDENCE LEDGER

The paper source includes EVIDENCE_LEDGER.md. Each quantitative manuscript statement is classified as verified, development-only, sealed, or pending. A verified entry names the producing command or log, source-data manifest, seed, checkpoint, and artifact fingerprint. Values without that chain are not printed as results. This policy is also why the earlier geometry, gravity, rate-equivalence, DFT-cosine, deployment, and phone-latency numbers are absent from this revision.